Mantle Cell Lymphoma

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87.1 Introduction

Mantle cell lymphoma (MCL) is an aggressive B-cell lymphoma, which is characterized by the chromosomal translocation t(11;14)(q13;q32) and overexpression of cyclin D1 in the vast majority of cases. Most patients present with advanced stage disease often with extra-nodal dissemination. High Ki67 proliferation index (Hoster et al. 2016), blastoid histologic variant (Bernard et al. 2001), and TP53 mutations represent high-risk features at diagnosis (Eskelund et al. 2017).

Front-line treatment of MCL has improved considerably, but success with chemotherapy-based treatments of relapsed and refractory (R/R) MCL has been limited (Dreyling et al. 2017). A paradigm shift from chemoimmunotherapy to targeted- and cellular therapies, such as chimeric antigen receptor (CAR) T-cell treatment, is currently changing the dismal prognosis of R/R MCL.

87.2 Autologous HCT

During recent years, the prognosis of patients with MCL has improved considerably, and the refinement of dose-intensified approaches such as auto-HCT has contributed significantly to this development. A prospective randomized trial by the European Mantle Cell Lymphoma Network (EMCLN) has demonstrated the superiority of auto-HCT consolidation over interferon maintenance (Dreyling et al. 2005) in the pre-RTX era. The introduction of RTX and the addition of high-dose cytarabine (HD-ARAC) to the induction treatment before auto-HCT have further improved PFS and OS of MCL patients (Cheah et al. 2016; Geisler et al. 2008). The benefit of HD-ARAC before auto-HCT could be confirmed in prospective clinical trial (Hermine et al. 2016). The French study group conducted a prospective randomized phase III trial (LyMa trial) that investigated RTX maintenance versus observation after auto-HCT in previously untreated MCL patients. Final results of the LyMa trial confirmed the superiority of RTX maintenance with regard to PFS and OS (Le Gouill et al. 2017). The beneficial effect of RTX maintenance was observed in both PET-positive and PET-negative patients after induction treatment prior to auto-HCT (Mei et al. 2017). This finding implies that the benefit of RTX maintenance after auto-HCT is present for low- and high-risk MCL patients (Dietrich et al. 2014b, Graf et al. 2015).

Very recently, first results of the TRIANGLE study revealed that the addition of Ibrutinib to HD-ARAC induction followed by RTX and ibrutinib maintenance further improved first-line treatment of MCL (Dreyling et al. 2022). Ibrutinib with RTX- and HD-ARAC-based induction treatment can therefore be considered as the new

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first-line standard treatment for young and fit MCL patients. A second experimental arm of the TRIANGLE study investigated ibrutinib and HD-ARAC induction treatment without auto-HCT consolidation. So far, it can be concluded that ibrutinib and HD-ARAC without auto-HCT is significantly better than HD-ARAC followed by auto-HCT. If the auto-HCT after ibrutinib and HD-ARAC induction can be omitted needs to be shown with longer follow-up of the TRIANLE study. In MIPI low-risk patients, however, it seems very unlikely that a consolidating auto-HCT further improves outcome. It seems also unlikely that *TP53*-mutated MCLs benefit from high-dose chemotherapy and auto-HCT. Auto-HCT in these two selected groups of MCL patients should therefore be critically discussed.

87.3 Chimeric Antigen Receptor T-Cell Therapy

Although a significant proportion of patients with MCL enjoy long-term disease control after intensive first-line therapies, relapse remains the main cause of treatment failure. The prognosis of patients with MCL recurrence after auto-HCT appears to be extremely poor, especially if occurring early after transplant (Dietrich et al. 2011, 2014a). A proportion of almost 40% of MCL patients relapsing after auto-HCT were reported to suffer from chemotherapy-refractory disease (Dietrich et al. 2014a) with a high prevalence of clonal TP53 mutations (Halldorsdottir et al. 2011).

The advance of autologous, CD19-directed CART-cell therapies marked a significant improvement for the treatment of R/R MCL. In patients with prior BTKi exposure, single infusion of Brexucabtagene autoleucel (brexu-cel) resulted in 91% response rate (68% CR) and median PFS of 28.2 month after three years of follow-up (Wang et al. 2023a). Real-world data from the Europe and United states confirmed the excellent outcome with significant proportion of long-term remissions (Iacoboni et al. 2022; Wang et al. 2023b).

Grade 3 or higher cytokine release syndromes (CRS) and neurologic events occurred in 15% and 31% of patients, respectively. Patients below and above the age of 65 years had similar outcomes after treatment with brexu-cel (Wang et al. 2023a), suggesting that CAR T-cell treatment can be applied in MCL patients who are not eligible for autologous or allogeneic HCT. Frail patients (ECOG $\geq$2), however, are more frequently affected by higher grade neurologic toxicities (ICANS) (Wang et al. 2023b). Although much better tolerable than allogeneic HCT cumulative 1-year nonrelapse mortality Brexucabtagene can reach 9.1% in patients with MCL.

Due to its better safety profile, CAR T-cell therapy can be considered as the new standard for MCL relapse after ibrutibib treatment. As a consequence, most MCL patients receive CAR T-cell therapy in third line or later, but there is interest in using CAR T-cell therapy in earlier lines. The CARMAN study conducted the European Mantle Cell Lymphoma Network (EMCLN) will investigate the efficacy of CAR T-cells in first line for high-risk patients.

87.4 Allogeneic HCT

In a large EBMT registry study, which investigated the outcome of MCL patients after first-line auto-HCT failure, 24% of all MCL patients received a rescue consolidation HCT. Only a minority of 2% received a second auto-HCT of whom only one patient experienced a long-term survival. These limited results do not justify a rescue auto-HCT as reasonable salvage strategy in this situation. In contrast, the majority of patients who received a second HCT underwent allo-HCT, and a significant proportion of them achieved a durable remission, translating into a 3-year OS of 43% (Dietrich et al. 2014a). Other registry studies and single-center experiences report similar results (Cook et al. 2010; Tam et al. 2009; Le Gouill et al. 2012).

Long-term efficacy of RIC allo-HCT was recently demonstrated in a large cohort of MCL patients (Robinson et al. 2018). The cumulative incidence of relapse was 40% at 5 years, and OS was 40% at 5 years. Patients who developed a chronic GVHD and/or patients who did not receive an in vivo TCD with CAMPATH had a

significantly lower relapse rate, suggesting the existence of a graft versus MCL effect. Despite long-term remissions after allo-HCT, chemorefractory disease (Robinson et al. 2018) or early relapse after first-line auto-HCT (Dietrich et al. 2014a) significantly reduced the long-term survival of MCL patients after allo-HCT. CAR-T-cell therapy seems to provide better outcome in this setting, although direct comparison is pending (Iacoboni et al. 2022; Wang et al. 2023a). It is important to note that there is a small group of relapsed MCL patients who survived longer than 5 years even without allo-HCT, suggesting a rather indolent disease course in a subset of patients (Dietrich et al. 2014a). Such patients with a low percentage of Ki67-positive tumor cells might not benefit from an allo-HCT.

Key Points

- Ibrutinib improves disease-free survival in addition to intensive induction- and RTX maintenance therapy.
- Auto-HCT in first remission should be questioned in low-risk patients and patients with *TP53* mutations not responding to chemotherapy.
- CD19-directed CAR T-cell therapy achieves high remission rates in relapsed and R/R MCL.
- Early treatment with CAR T-cells is currently evaluated by EMCLN in the CARMAN trial.
- While curative potential of CART-cell therapy in MCL remains unclear, allo-HCT should be evaluated for MCL patients relapsing after CAR T-cell therapy.

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